16 April 2026 09:06

Loops:-

Repeat the block of code multiple times without writing it again and again .

ATM :-

Please choose the option :-

1. **Withdrawal**
2. **Balance enquiry**
3. **Pin change**
4. **Mini statement**

Enter valid input:
5

Withdraw(amount)
{
Pin_check()
{
Pin=1234
If(pin==1234)

Balance=5000;
If(amount<=5000)
    {
    Balance=Balance-amount
    }
Else
{
Print (insufficient funds)
}
}
}
Else wrong pin

For loop, while loop :-
For loop:- iterate over a sequence (list, range, string)
While loop:- keep running as long as condition is True

For Loop:-
Syntax-: for variable in sequence :

```
# block of code
for number in range(1,11):
    print(number)            number=1,2,3 11
```

```
While loop:

Syntax:- while condition:
          #Block of code
```

```
count=1
while count<=5:
    print(count)
    count +=1
```

Count	Condition	Assignment
1	1<=5 t	Count=2
2	2<=5 T	Count=3
3	3<=5 T	Count=4
4	4<=5 T	Count=5
5	5<=5 t	Count=6
6	6<=5 F	

Note : - If condition never becomes false --> infinite loop

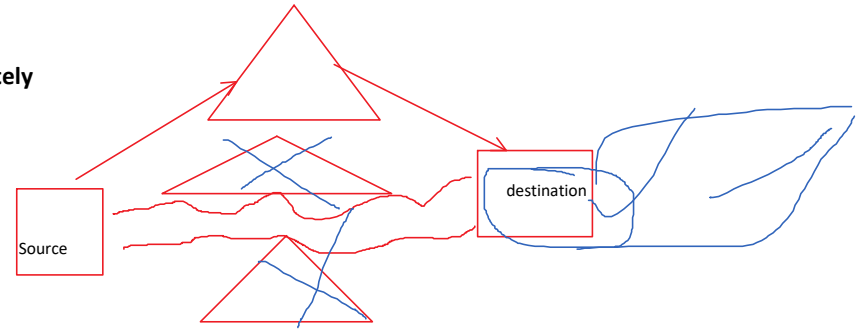
Loop control statements:

1. **break**

2. **Continue**
3. **pass**

Break:-stop the flow of execution / exits loop completely

```
for number in range(1,11):
     if number == 5:
       # print(number)
        break
     print(number)
```



Continue:-Skip current execution

```
for number in range(5):
{
    if number == 2:
        continue   execution stops
    print(number)
}
```

Number	Condition	Result
0	0==2 F	0
1	1==2 F	1
2	2==2 T	
3	3==2 F	3
4	4==2 F	4

PASS - DOES NOTHING (PLACEHOLDER)

Nested Loops :
Loop inside another loop

For {

for {

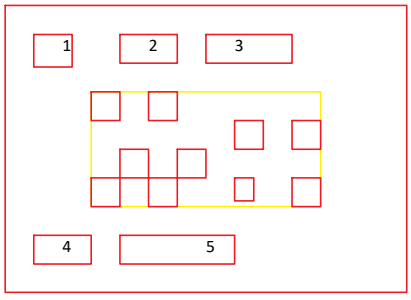
}

}

- once the flow of execution move inside the loop , the inner loop starts the execution , and the inner loop continues the execution until the loop finished , once the inner loop finished the execution then only the outer loop starts the execution .

Outer loop repeats until the sequence finished

But the inner loop keep on iterating how many number of iterations can be done by outer loop



0 TO 2
0,1

3 --0-->2

```
for i in range(3):
    for j in range(2):
        print(i,j)
```

Outer loop	I
IST	0
2ND	1
3RD	2

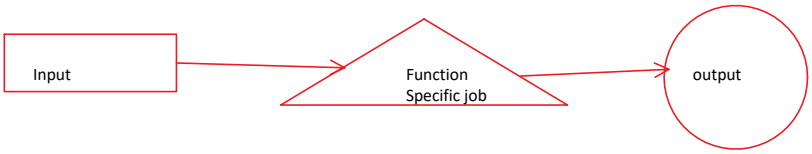
Inner loop	J
IST ITERATION	0
	1
2ND ITERATION	0
	1
3RD	0
	1

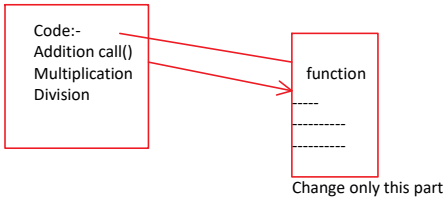
OUTPUT

I	J
0	0
0	1
1	0
1	1
2	0
2	1

Functions :-

Function is a named , reusable block of code that performs a specific task , instead of writing same code in to multiple times .





Easy to change
Fast change
Safe (low risk)

Types of Function:-

1. Built in function, print(),len(),str(),int(),input(),type()...
2. User defined function
3. Lambda function
4. Standard libraries (math, date , random)
5. Recursive function (factorial)

User defined function
Write by own
1. **Define**
2. **Use (call)**

Function definition :

```
def function_name( .....):
    <line of code>
    <line of code>
    <line of code>
    <line of code>
    <line of code>
```

Parameters

Description
declaration

body

Function call:
function_name()

Execution Arguments

```
    print("start")
def greet():
    print("heloo")
greet() call
print ("end")
```

Memory
Place holder
Greet
Print("hello)

Output

Start
Heloo
end

No input /output

Only input - there is no output returns
Input and output - we are expecting ouput
Parameters:-
Names used in function definition that describe what data the function expects
Arguments
Actual values passed in a function call that are assigned to parameters

Multiple input and outputs ...

```
def factorial(n):
    if n ==0 or n == 1:
        return 1
     #recursive step  n!= n * (n-1)
    return n * factorial(n -1) 4 *3 *2 *1
number=4
result=factorial(number)
```

```
print(f"the factorial of {number} is : {result}")
```

```
Naming rules :- creating user defined functions:
1. The function name must start with letter or _
2. Use meaningful names
   Def calucualte_salary() ---fine
   Def cs()-- bad
3. Indentation
  a. Def test():
    i. Print()
4. Avoid global variables
```

Module?

Modules is a python file which contains python code(reusable) like functions, classes , variables	

It is saved with .py

Why use modules?

- Reusability
- Code organization
- Easy debugging
- Save time
- Collaboration-friendly
- Reduce duplication

1. Built-in module
2. User defined

Builtin modules:
How to import :

3. Import module name : import complete module : import math -->math.sqrt(20)
4. From module import function names :from math import sqrt,pi --> sqrt()
5. From module import * : not recommended --> sin(1)
6. Import module as alias : import datetime as dt

User defined modules :- modules can be created by user

A package is a folder contains multiple moduels , usually with __init__.py fine